

Web Scraping and Data Collection for Life and Social Sciences

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Abstract

The increasing availability of online records, public registries, organizational websites, institutional repositories, and platform-based information has transformed empirical research in the life and social sciences. Web scraping and related forms of web-based data collection now play a central role in the construction of datasets for descriptive inquiry, exploratory analysis, measurement, and subsequent inferential work. Yet the methodological value of these materials depends less on the mere act of extraction than on principled source selection, careful definition of the unit of analysis, transparent transformation of web content into tabular structures, and credible validation of the resulting data. This paper situates web scraping within a broader research workflow for digital data construction. It reviews relevant methodological literature on computational social science, text-as-data, digital trace collection, datafication, platform governance, and internet research ethics; discusses the opportunities and limitations of web-based sources for the life and social sciences; examines key problems of representativeness, structural bias, standardization, validation, and reproducibility; and proposes a pedagogically oriented framework for introducing researchers to web-based data collection in a no-code environment. Restricting the computational discussion to a single penultimate section, the paper outlines a workflow in which Python and R scripts hosted in a public repository support an interface-based application through which participants can upload URLs or datasets, configure source types and units of analysis, apply cleaning rules, and generate preliminary descriptive outputs without writing code. The argument is that web scraping is best understood and taught as one component of data construction for substantive research questions in health, life, and social sciences, and that no-code environments are most defensible in introductory and pedagogical contexts where conceptual transparency can be foregrounded without claiming full procedural transparency.

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1 Introduction

Digital sources have become indispensable to contemporary empirical research. Public institutions, professional organizations, laboratories, government agencies, legislative bodies, clinical trial registries, funding portals, directories, and archival collections increasingly publish structured and semi-structured information online. For researchers in the life and social sciences, the web is therefore not merely a communication medium; it is also a field site, an archive, and a distributed infrastructure for the production of observable records. The methodological consequence is straightforward: if research increasingly depends on online information, then the collection, organization, and evaluation of web-based data must be treated as a serious part of research design rather than as an informal preliminary task.

The emergence of computational social science made this point early by arguing that digital environments generate new forms of social data at scale and create unprecedented opportunities for measurement and analysis [Lazer et al., 2009, 2020]. More recent work

has shown that the central challenge is not simply the availability of digital traces but the capacity to align those traces with meaningful research questions, ethically acceptable collection strategies, and defensible validation procedures [Grimmer et al., 2022, Ohme et al., 2024, Brown et al., 2025]. At the same time, scholarship on datafication and platform society has emphasized that digital records are shaped by socio-technical infrastructures, institutional incentives, and platform governance arrangements rather than appearing as neutral by-products of online activity [Kitchin, 2014, van Dijck et al., 2018]. This broader framing is especially relevant to web scraping, because the researcher does not encounter raw social reality on the web, but already formatted, categorized, and hierarchically organized representations.

In adjacent work on text analysis, Grimmer, Roberts, and Stewart emphasize that data construction is inseparable from theory, representation, and validation; there is no value-free corpus construction, and the suitability of a method depends on the task at hand [Grimmer et al., 2022]. Those insights are highly useful for thinking about web-derived data, but they should not be transferred uncritically. The movement from text-as-data to web scraping is analogical rather than strictly equivalent. Corpus construction in text analysis and structured extraction from web pages both involve selection, representation, and validation, yet they do not pose identical technical problems. In web scraping, the researcher must often navigate Document Object Model (DOM) hierarchies, pagination schemes, hidden fields, and changing interface logic, all of which affect what can be observed and recorded. The present paper therefore borrows from the text-as-data literature selectively, while treating web extraction as a distinct measurement problem.

This paper develops that argument in the context of web scraping and digital data collection for the life and social sciences. The aim is not to provide a comprehensive technical manual on scraping tools. Nor is it to defend an unrestricted expansion of data extraction from any online source. Instead, the goal is to provide a methodologically grounded account of how web scraping can be understood as one stage within a broader research workflow. That workflow begins with a substantive question, proceeds through source identification and unit definition, and continues through extraction, cleaning, documentation, validation, preliminary exploration, and interpretation. Understood in these terms, web scraping is less a narrow programming trick than a practical and epistemic bridge between online information environments and research-ready datasets.

The paper also has a pedagogical purpose. In many academic settings, especially in professional training for early-career scholars and advanced doctoral students, the demand for digital methods exceeds the coding experience of participants. This gap has important implications for how web-based data collection is best introduced. If a seminar or workshop promises to cover the whole universe of scraping techniques in a short time, it is likely to be unrealistic. If, however, instruction is designed around a carefully delimited no-code workflow built on open scripts and an interface-based application, then participants can learn

the logic of digital data construction without being forced prematurely into programming. For that reason, the programmatic and no-code dimension of this paper is reserved for a single section immediately preceding the conclusion. The earlier sections first establish why web-based data collection matters methodologically and substantively.

The remainder of the paper proceeds as follows. Section 2 reviews the literature on computational social science, text-as-data, digital trace data, datafication, and the methodological status of web scraping. Section 3 discusses what kinds of web sources are especially relevant for researchers in life and social sciences. Section 4 examines core design principles: source selection, unit definition, field construction, measurement, and bias. Section 5 addresses validation, data quality, and ethical issues. Section 6 considers applications to health, life, and social science research. Section 7 presents a no-code computational workflow based on repository-hosted Python and R scripts and an interface-based application. Section 8 concludes.

2 Literature Review: Web-Based Data in Computational Research

The methodological literature relevant to web scraping spans several adjacent fields rather than a single unified tradition. In computational social science, online environments are important because they generate records of interaction, communication, institutional activity, and organizational presence that can be measured and analyzed at scales previously difficult to reach [Lazer et al., 2009, 2020]. This literature often emphasizes the opportunity structure created by digital platforms, but it also warns that new forms of data introduce new forms of bias, access inequality, and interpretive difficulty. The abundance of observable traces does not eliminate the need for careful research design; it often makes that need more urgent.

A second body of work concerns the methodological treatment of texts and digital records as data. Grimmer, Roberts, and Stewart argue that computational methods do not replace social science reasoning but extend it by allowing researchers to transform large, complex, and often unstructured materials into analyzable representations [Grimmer et al., 2022]. Their six principles of text analysis are especially relevant to web scraping at the level of research design: theory remains indispensable, computational tools should augment rather than replace human judgment, the development of valid measures is iterative, representational choices matter, no method is universally best, and validation is essential. Even so, the relationship should be described carefully. Web scraping is not simply text analysis by other means. The representational challenge of scraping frequently lies in converting DOM structures, nested page elements, dynamically loaded content, or repeated templates into analyzable rows and columns. The text-as-data framework is therefore a valuable analogy for measurement and validation, not a complete blueprint for web extraction.

A third line of scholarship deals more directly with web-derived and digital trace data. Li, Zhou, and Cai present three cases of collecting web information for social science research and show that the technical means of collection must be linked to the structure of the target source and the substantive purpose of the study [Li et al., 2021]. Their discussion is valuable because it resists the idea that web collection is a generic operation. Different web sources embed different units, hierarchies, and regularities, and the data collection strategy should be adapted accordingly. Similarly, Ohme and coauthors situate web-based data collection within a broader discussion of digital trace data, emphasizing access modes such as APIs, data donation, and screen tracking, while underscoring the importance of privacy, validity, and fit between research question and collection method [Ohme et al., 2024]. Their account is also useful because it highlights methodological uncertainty: digital trace sources are rarely created for research purposes and may change unexpectedly, thereby affecting completeness, reproducibility, and interpretability.

A fourth literature concerns the socio-technical constitution of digital data. Kitchen argues that data are not simply found but generated within particular infrastructures, classificatory systems, and technical regimes [Kitchen, 2014]. Van Dijck, Poell, and de Waal similarly describe digital platforms as institutional and infrastructural environments that shape what becomes visible, actionable, and exchangeable as data [van Dijck et al., 2018]. This perspective is important for web scraping because it situates collection within broader questions of platform governance and infrastructural mediation. A website does not merely host information; it presents information according to page architectures, menus, templates, filters, visibility rules, and update practices that affect what the scraper can observe.

A fifth literature addresses legal and ethical considerations. Brown and colleagues argue that web scraping for research cannot be understood solely as a technical procedure. It also raises legal, institutional, scientific, and ethical questions about data access, platform governance, consent, privacy, and responsible use [Brown et al., 2025]. Their analysis is particularly helpful because it distinguishes different legal dimensions. Terms of Service restrictions often raise contractual questions, whereas copyright and fair use concerns arise from statutory regimes; these are related but not identical issues [Brown et al., 2025]. Luscombe, Dick, and Walby similarly describe web scraping in the social sciences as requiring algorithmic thinking in the public interest, with attention to technical feasibility, legal constraints, and ethical judgment [Luscombe et al., 2022]. These contributions are especially important because they counter the misconception that publicly accessible online data are automatically free of ethical concern.

Finally, there is a pedagogical literature that helps frame how researchers and students should be introduced to scraping. Dogucu and Dogucu and Çetinkaya-Rundel [2021] argue that web scraping belongs in the statistics and data science curriculum because it exposes learners to data acquisition, irregular structures, reproducibility, and the critical evaluation of web-based sources. Their argument has a broader implication for professional methods

training. Teaching scraping effectively means teaching not only extraction commands but also the logic of transforming heterogeneous online content into analyzable evidence.

Taken together, these literatures support a view of web scraping as a component of research design rather than a stand-alone technical skill. They also suggest that digital data collection should be judged not only by whether it can be done, but by whether the resulting data are suited to the substantive and inferential goals of a project.

3 Web Sources as Research Material in Life and Social Sciences

The practical appeal of web scraping in life and social sciences derives from the diversity of online sources that can be incorporated into research workflows. In the health sciences, one obvious category consists of public registries and structured informational portals. Clinical trial registries, hospital and provider directories, public health dashboards, policy repositories, professional association pages, and listings of grant programs all contain information that may be relevant for mapping institutional activity, identifying populations of organizations, or building descriptive datasets for exploratory analysis. These sources are especially attractive because they are often public-facing, regularly updated, and already organized around recurring units such as trial entries, institutions, programs, or locations.

In the life sciences, researchers may encounter comparable opportunities in institutional laboratory pages, funding databases, conference or program directories, repositories of research centers, open-access registries of species or environmental observations, and publicly posted descriptions of projects and collaborations. Even when these sources are not designed as research datasets, they often contain repeated structures that can be reconfigured into rows and variables. The methodological task is to determine whether those structures correspond closely enough to the research unit of interest to justify extraction.

In the social sciences, the range of applicable sources is especially broad. Legislative records, policy portals, court or administrative listings, organizational websites, public directories, campaign pages, municipal repositories, and institutional announcements can all serve as starting points for structured data collection. Some projects may focus on institutions, others on events, others on actors or organizational fields. In each case, the web source offers a practical means of constructing a dataset where no prepackaged table exists.

The availability of such sources should not be confused with methodological simplicity. A web page is not equivalent to a variable, and a collection of URLs is not equivalent to a dataset. The same site may mix textual, tabular, and navigational information; may display different information for different units; or may omit critical fields in irregular ways. The researcher must therefore ask not only whether data can be extracted, but what exactly a unit on the page represents, how stable the source structure is, and whether the resulting

records can support the planned analysis. This is where a source-oriented view of research design becomes essential.

The book by Grimmer et al. [2022] is helpful here, even though its focus is text rather than websites as such. Their discussion of corpus construction and selection bias makes clear that data collection is always shaped by retrieval decisions, representational choices, and substantive theory. A similar point applies to web sources, but only with appropriate caution. Websites are not neutral containers of information, and neither are they reducible to corpora. They are produced by institutions with their own incentives, maintenance practices, interface designs, and communicative purposes. A legislative portal may reflect official classifications that are useful for some questions and misleading for others. A directory may include only registered entities. A clinical registry may vary in completeness across regions or periods. These characteristics are not defects in themselves, but they must be understood if web-derived data are to be used responsibly.

4 Research Design Principles for Web Scraping and Data Collection

4.1 From research question to source selection

A robust web-based data collection project begins with a research question rather than with an available website. This may sound obvious, but the abundance of accessible online information often tempts researchers into source-led rather than question-led design. When that happens, extraction precedes conceptualization, and the resulting dataset may be rich in fields but weak in interpretive value. The literature on computational social research consistently warns against this inversion. As Grimmer et al. [2022] emphasize, methods are useful only in relation to tasks, and tasks derive from substantive inquiry. Likewise, Lazer et al. [2020] note that the promise of digital data depends on integrating technical capability with social scientific reasoning.

Question-led design affects source selection in several ways. First, it determines what population or domain the source should represent. If the goal is to study clinical trials on a certain condition, then the relevant source may be a registry organized around trial entries. If the goal is to map legislative activity, then a public portal organized around bills or sessions may be more suitable. Second, it determines which fields matter. A source that offers rich descriptive text but lacks stable dates or identifiers may be unsuitable if temporal comparison is central to the project. Third, it determines the acceptable level of incompleteness or noise. A source that is imperfect may still be useful for exploratory mapping, but not for strong inferential claims.

The practical implication is that web scraping should begin with a documented ratio-

nale for source choice. Why this source rather than another? What population does it approximate? What unit of observation does it naturally expose? What known exclusions or irregularities might affect interpretation? These questions help transform extraction from a technical convenience into a reproducible research decision.

4.2 Defining the unit of analysis

One of the most consequential steps in web-based data construction is defining the unit of analysis. Web pages often contain nested structures: a page may represent an institution, while sub-elements within the page represent programs, persons, events, or locations. A directory page may list organizations, each of which links to further profile pages. A legislative portal may organize materials by session, bill, committee, or vote. Without explicit unit definition, a researcher can easily mix levels of analysis or produce a dataset whose rows do not correspond consistently to a meaningful entity.

The issue has analogues in text analysis, where choices about the unit of document representation affect downstream measures and inferences [Grimmer et al., 2022]. In web scraping, however, the problem is compounded by page architecture. A row in the final table is usually not given by the source itself; it is produced by translating HTML structure into a tabular schema. A clear statement of what one row in the final dataset represents is therefore indispensable. Depending on the project, a row might correspond to a clinical trial, an institution, a legislative item, a laboratory, a program, a funding call, or an event. That decision should precede field extraction and guide subsequent cleaning.

4.3 Field construction, measurement, and schema design

Once the unit of analysis is defined, the next task is to identify the fields that will represent each unit in a dataset. Here again, the act of scraping is only the beginning. Most web-sites were not designed with a research schema in mind. Relevant information may appear under inconsistent labels, in mixed text blocks, across multiple pages, or behind interactive elements. A common mistake is to collect everything that can be extracted and postpone structure until later. While exhaustive extraction can occasionally be useful, it often creates cumbersome datasets that obscure rather than clarify the substantive object.

A more disciplined approach begins with a provisional schema: title, identifier, date, institution, category, location, URL, and notes are examples of generic fields that often recur across domains. This schema can then be adapted to the source. In some cases, a field must be derived from multiple page elements; in others, a field must be harmonized across sources. The key point is that field construction is a measurement decision. It defines what counts as an observation and what form of comparability the resulting table will support.

Scraping is therefore not only extraction but also a model of source structure. When

researchers select CSS selectors, XPath expressions, table boundaries, pagination rules, or click paths, they introduce a measurement layer that mediates between the website and the dataset. DOM structure is not equivalent to the ontology of the phenomenon being studied. Pagination, dynamic loading, hidden fields, collapsible sections, and inconsistent markup all shape selective observability. What becomes visible in the final dataset is partly a function of substantive source content and partly a function of how the extraction logic models that content. Recognizing this measurement layer is essential for responsible interpretation.

4.4 Bias, coverage, and comparability

All found data raise questions of bias and comparability, and web-based data are no exception. Grimmer, Roberts, and Stewart distinguish several forms of bias in corpus construction, including resource, incentive, medium, and retrieval bias [Grimmer et al., 2022]. These distinctions can be translated productively into the context of web scraping, but only as an analogy rather than a strict extension. Resource bias appears when some institutions maintain richer or more accessible web pages than others. Incentive bias appears when actors selectively disclose information for strategic reasons. Medium bias arises because websites privilege some kinds of information and omit others. Retrieval bias appears when the researcher’s collection procedure captures only pages with certain access paths or levels of search visibility.

Web scraping also introduces an additional category that deserves explicit emphasis: structural bias, or DOM-dependent extraction bias. This occurs when page architecture or interface design systematically affects what is retrievable. Information that appears only after clicking, scrolling, or expanding elements may be excluded. Fields embedded in scripts or dynamically loaded components may be missed. Layout irregularities may cause some records to parse cleanly while others fail silently. Structural bias matters because it does not arise solely from the social process that generated the source; it also arises from the interaction between source structure and extraction logic.

These biases affect not only who or what enters the dataset, but also which variables are observed with relative completeness. A clinical registry may be comprehensive in one jurisdiction and partial in another. A legislative site may preserve official actions while omitting informal deliberation. A university directory may include current personnel but not historical entries. Comparability is therefore never automatic. It must be argued, tested where possible, and documented.

5 Validation, Data Quality, and Ethics

5.1 Why validation is central

Validation is among the most important themes shared by the literatures on text-as-data and digital trace research. For Grimmer et al. [2022], validation is not an optional afterthought but a constitutive part of computational measurement. The same argument applies to web-derived data. A dataset can be technically well formed and yet conceptually weak. The fact that rows and columns are complete does not by itself show that the dataset measures what the researcher intends.

Validation in web scraping has at least five dimensions. The first is structural validation: confirming that the extraction process captured the intended units and fields consistently across pages. The second is semantic validation: checking that the values entered in a field mean the same thing across observations. The third is coverage validation: assessing whether the source population and the extracted sample align sufficiently for the research purpose. The fourth is cross-source validation: comparing extracted records against external benchmark datasets or triangulating with other scraped or manually collected sources wherever feasible. The fifth is analytic validation: determining whether the resulting data behave plausibly in summaries, distributions, or cross-checks with outside information.

These dimensions often require iteration. Researchers may need to inspect random samples of extracted records, compare outputs against source pages, triangulate with manually collected benchmarks, or assess how stable key fields remain across repeated collection attempts. They may also need to revise field definitions or cleaning rules once inconsistencies become visible. Such iteration is not a sign of methodological weakness; it is part of the normal process of turning found digital information into evidence.

5.2 Cleaning, standardization, and documentation

Web-based data are rarely analysis-ready at the moment of extraction. Dates may appear in multiple formats, names may contain inconsistent whitespace or punctuation, labels may vary across source pages, and supposedly identical categories may be spelled in slightly different ways. Duplicate records may arise when directories display the same entity through multiple paths. Missing values may signal absent information, extraction failure, or source-specific reporting norms. Cleaning therefore requires both procedural regularity and contextual judgment.

The literature on teaching web scraping usefully emphasizes this point. Dogucu and Çetinkaya-Rundel [2021] show that data acquisition from the web naturally introduces students to the realities of messy data and to the importance of reproducible preparation. In research settings, cleaning should not be understood as cosmetic improvement but as an explicit transformation of raw outputs into a more interpretable and comparable form.

That transformation should be documented. Processing logs, metadata sheets, preserved versions of raw and cleaned files, and explicit notes on parsing assumptions help maintain transparency and support later replication.

5.3 Ethical and institutional considerations

The ethical status of web scraping has become increasingly central to methodological discussion. Brown and colleagues argue that researchers must consider not only legality but also scientific integrity, institutional review, platform expectations, privacy, and the potential downstream risks associated with collection and reuse [Brown et al., 2025]. Luscombe, Dick, and Walby similarly stress that web scraping in the social sciences involves navigating technical, legal, and ethical hurdles rather than merely solving engineering problems [Luscombe et al., 2022]. These arguments are especially important in academic environments, where the mere public accessibility of a web page may create a false sense that no ethical reflection is required.

Legal and ethical questions should also be distinguished more carefully than they often are in practice. Terms of Service restrictions frequently raise contractual questions about permitted use, assent, and enforceability, whereas copyright, database rights, privacy, and fair use involve statutory or regulatory domains [Brown et al., 2025]. These considerations may overlap in specific cases, but they are not conceptually identical. For researchers, this means that a project can be ethically questionable without being plainly unlawful, and it can raise legal uncertainty even when the collected material appears publicly accessible.

Ethical judgment in this domain cannot be reduced to a mechanical rule. Context matters. Some sources are clearly public, institutional, and intended for wide circulation. Others contain information that is publicly accessible yet contextually sensitive. Researchers must consider whether scraping changes the scale or visibility of data use, whether individuals are identifiable, whether the source has terms or technical measures that bear on access, and whether storing or redistributing the extracted data could create risks beyond the original context of publication. Ethical risk arises not only from access, but also from scale transformation. Aggregating contextually public information into a decontextualized large-scale dataset can create new forms of visibility, facilitate re-identification, or enable downstream uses far removed from the conditions under which the material originally appeared.

The broader internet research ethics literature reinforces this emphasis on contextual decision making rather than one-size-fits-all prescriptions [Ess and Jones, 2004]. This is consistent with the more recent call by Brown et al. [2025] for integrating ethical reflection with scientific and institutional considerations. For life and social science research, that means building ethics into design discussions from the outset: at source selection, at data storage, at reporting, and at decisions about what should be shared publicly after collection.

6 Applications to Health, Life, and Social Science Research

A key advantage of web-based data collection is that it can support substantively diverse projects while relying on a common methodological logic. In health sciences, for example, clinical trial registries can be used to construct datasets on trial characteristics, geographic distribution, sponsor types, or recruitment status. Public directories of providers, institutions, or laboratories can support descriptive mapping of organizational fields. Policy portals can be used to track public health directives or institutional responses across jurisdictions. These applications are particularly useful for exploratory work, scoping studies, and descriptive analyses that precede more formal modeling, but their value depends on how carefully registry definitions, update cycles, and jurisdictional differences are handled.

In the life sciences, web-derived data can help researchers map laboratories, centers, projects, funding programs, or collaborative infrastructures. Institutional and funding websites often contain recurrent fields such as investigator names, research areas, sponsoring agencies, deadlines, or geographic location. When organized carefully, these materials can support the study of research ecosystems, opportunity structures, and patterns of institutional specialization. They can also serve as scaffolding for later linkage to bibliometric, geographic, or administrative data. The analytical payoff is greatest when the researcher can explain what the web source does and does not capture about the underlying scientific field.

In the social sciences, the possibilities are broader still. Legislative records can support analyses of bill characteristics, sponsors, committee assignments, or issue classifications. Public organizational websites can be used to map advocacy groups, firms, nonprofits, or educational institutions. Government and municipal portals can support work on procurement, service provision, regulation, or policy diffusion. Event pages, public announcements, or directory structures can be transformed into datasets on programs, participation opportunities, or institutional fields. Yet these uses should not be treated as interchangeable. A legislative portal, for instance, may be strong for official procedural records but weak for informal negotiations; a directory may reveal organizational visibility rather than organizational capacity.

What these examples share is not a common substantive domain but a common workflow of data construction. In each case, the researcher begins with a source that contains repeated web structures and asks whether those structures can be aligned to a meaningful unit of analysis. Extraction is followed by schema design, cleaning, validation, and preliminary description. This commonality is what makes web scraping teachable across disciplines. The method is not tied to a single subject matter, but it does require domain-sensitive examples to make the workflow concrete and analytically credible.

The literature on digital trace data is instructive here. Ohme and colleagues argue that

researchers should choose among APIs, donation-based approaches, tracking, and related forms of digital trace collection according to the substantive object and the mode of access that best fits it [Ohme et al., 2024]. Web scraping belongs within this broader menu rather than standing outside it. In some projects, scraping may be the only viable route to a structured dataset; in others, it may be inferior to an API, an official bulk download, or a benchmark administrative source. The methodological point is not that scraping is universally preferable, but that it can be highly valuable when matched carefully to the question, source, and validation strategy.

7 Technical Failure Modes in Web-Based Data Collection

The methodological discussion of web scraping often emphasizes research design, validation, and ethical considerations. However, in practice, a substantial share of error arises from technical failure modes that do not necessarily produce explicit warnings but nonetheless affect the integrity of the resulting dataset. Recognizing these failure modes is essential for assessing data quality and for designing robust collection procedures.

Websites are not static objects. Their underlying HTML structure may change over time due to redesigns, content management updates, or A/B testing practices. Scraping procedures that rely on fixed selectors or paths may therefore fail silently when the structure shifts. Even minor changes in class names, tag hierarchies, or pagination logic can lead to partial or incorrect extraction. As a result, reproducibility requires not only preserving code but also documenting the temporal context of data collection and, where possible, archiving source pages.

In addition, many contemporary websites rely on client-side rendering through JavaScript, which implies that the content visible in a browser may not be present in the initial HTML response. In such cases, simple HTTP requests may return incomplete data. Researchers must therefore distinguish between static and dynamically rendered content and select appropriate tools, such as headless browsers or API-based access, when necessary. Failure to do so may result in systematically missing fields or observations [Mitchell, 2018].

Web servers may also impose constraints on automated access, including rate limiting, CAPTCHA challenges, authentication requirements, or blocking of repeated requests. These mechanisms are designed to protect infrastructure and enforce platform policies. From a research perspective, they introduce both technical and methodological complications. Technically, they may lead to incomplete data collection if requests are throttled or denied. Methodologically, they may induce selection bias if only a subset of pages can be accessed under given constraints [Lazer et al., 2020, Brown et al., 2025].

Even when extraction succeeds at the level of retrieving content, the mapping from web

elements to structured fields is often ambiguous. Identical labels may refer to different concepts across pages, while semantically equivalent information may appear under different labels or formats. Date fields may vary in representation, organizational names may include inconsistent abbreviations, and categories may not be standardized across entries. These issues complicate schema construction and may introduce measurement error if not addressed through systematic cleaning and validation.

One of the most consequential risks in web scraping is silent partial extraction. A script may execute without error while capturing only a subset of intended observations. This can occur due to incomplete pagination, hidden content, conditional loading of elements, or interruptions in the request process. Because no explicit failure is reported, such omissions may remain undetected unless specific validation checks are implemented. Comparing counts against expected totals, sampling source pages for verification, and logging extraction coverage are therefore critical practices.

Taken together, these technical failure modes reinforce the broader argument that web scraping is not merely a mechanical extraction step but a component of measurement. The structure of the source, the mode of access, and the stability of the extraction procedure jointly determine what is observed and what remains unobserved. Incorporating awareness of these risks into research design requires explicit documentation of collection procedures, iterative validation, and, where feasible, triangulation with alternative sources. In this sense, technical robustness and methodological validity are closely intertwined.

8 A No-Code Computational Workflow for Research Training

The previous sections have argued that web scraping is best understood as one stage within a broader process of digital data construction. This section turns to the pedagogical and computational question of how such a workflow can be taught and implemented for researchers with little or no programming background. In keeping with the structure of this paper, the discussion of programmatic and no-code matters is confined to this single penultimate section.

The central premise is that a no-code seminar or training environment should not hide the underlying logic of data collection. Rather, it should separate conceptual learning from the immediate burden of writing code. One practical way to do so is to host the underlying computational procedures in a public repository containing parallel scripts in Python and R, while exposing those procedures to participants through an interface-based application. The repository provides inspectable procedures, reproducible versioning, and future extensibility; the application provides accessibility. Even so, the transparency provided by such an arrangement is conceptual rather than fully procedural. Participants can understand

the workflow, the sequence of operations, and the meaning of key design choices without necessarily seeing every implementation detail required for a full replication audit.

A pedagogically sound application should be modular. Instead of forcing all users into a single start point, it should allow entry from different stages of the workflow. Some users may begin with a single URL. Others may upload a list of URLs. Others may already possess a spreadsheet or a messy dataset collected elsewhere and require only cleaning and preliminary exploration. A modular architecture therefore fits both the diversity of participant needs and the methodological principle that data work is iterative.

A useful interface can be organized around five modules. The first module allows users to choose the input type: single URL, list of URLs, CSV or Excel file with target links, or preexisting dataset. The second module handles collection by offering source-type choices such as listing page, profile structure, directory, table, or already tabular file. The third module addresses cleaning and restructuring, including duplicate removal, field renaming, date standardization, whitespace trimming, category harmonization, column splitting, and exclusion of empty records. The fourth module supports preliminary exploration through summary tables, missingness checks, simple distributions, and basic visual displays. The fifth module enables export of cleaned datasets, metadata sheets, and processing summaries.

The most important feature of such an application is not automation alone but configurable transparency. Participants should be able to define the unit of analysis, map their fields to a generic schema, choose which cleaning rules to apply, preview the effects of transformations, and document the resulting output. These interface choices translate methodological decisions into accessible controls. A participant collecting legislative records might define each row as a bill and map fields to identifier, session, date, sponsor, committee, and status. A participant working with a clinical registry might define each row as a trial and map fields to trial ID, sponsor, condition, location, phase, and status. Another participant may skip collection entirely and use the same interface to standardize an already existing spreadsheet.

The role of the repository remains essential, but so do its limits. Hosting both Python and R scripts serves several purposes. It shows that the no-code interface is grounded in inspectable procedures rather than opaque manipulation. It supports reproducibility by preserving the logic of extraction and cleaning. It creates opportunities for later extension by more technically experienced collaborators. However, repository-backed scraping workflows are also fragile. Websites change, HTML structures shift, and interface elements are redesigned. A no-code user may understand the conceptual workflow perfectly well and still be unable to diagnose a broken selector, a failed parser, or an altered pagination rule. For that reason, a no-code architecture is best viewed as appropriate for introductory and pedagogical contexts, where the aim is to make the logic of digital data construction accessible without claiming that the interface alone can eliminate the need for technical maintenance.

The no-code approach is therefore not a substitute for all forms of methodological trans-

parency. Rather, it can foreground core design choices more clearly than a code-first workshop for beginners. Participants can concentrate on source suitability, unit definition, field mapping, validation, and ethics while still seeing that each operation corresponds to explicit computational steps. In that limited but important sense, a carefully designed no-code environment may offer a realistic entry point for early-career researchers in the life and social sciences.

The implementation of a no-code environment for web-based research introduces specific challenges regarding the long-term viability of the data construction process, particularly because web sources are not static containers but are products of institutions with evolving maintenance practices [Grimmer et al., 2022]. Unlike manual data entry, web-based collection relies on the structural stability of the target source, which is frequently subject to change [Ohme et al., 2024]. While a no-code interface lowers the entry barrier for researchers, it creates a significant technical dependency on the underlying Python and R scripts hosted in public repositories.

As Dogucu and Çetinkaya-Rundel [2021] argue, teaching scraping effectively means moving beyond simple extraction commands to include the logic of transforming heterogeneous online content into analyzable evidence. To ensure scientific integrity and transparency, the workflow must support the preservation of raw and cleaned file versions alongside processing logs [Brown et al., 2025]. Maintenance involves a continuous cycle of semantic and structural validation to ensure that field definitions remain consistent as source structures evolve. Ultimately, the value of digital trace data depends on the capacity to align these technical traces with ethically acceptable strategies and defensible validation procedures over time [Ohme et al., 2024, Brown et al., 2025].

9 Conclusion

Web scraping and web-based data collection now occupy an important place in empirical research across the life and social sciences. Their importance does not stem merely from technical novelty, but from the fact that an increasing share of relevant information is published, updated, and structured online. Public registries, directories, legislative portals, organizational websites, and institutional repositories provide researchers with opportunities to construct datasets that were once difficult or impossible to assemble. Yet the methodological value of those opportunities depends on design choices that precede and follow extraction itself.

This paper has argued that web scraping is best understood as part of a broader workflow of data construction. That workflow begins with a substantive question, continues through source selection and unit definition, and requires careful field construction, recognition of scraping as a measurement layer, cleaning, validation, and ethical reflection. The relevant literatures in computational social science, text-as-data, digital trace research, datafication,

platform governance, and internet research ethics all support this interpretation. They converge on the point that computational access to online materials does not reduce the importance of theory, measurement, or validation; it amplifies it.

For this reason, the pedagogical challenge is not to promise exhaustive technical coverage in a short training format. It is to provide researchers with a realistic and transparent way to understand how web-based information can be turned into structured, inspectable, and reusable research material. A no-code workflow built on open Python and R scripts and implemented through an interface-based application offers one plausible approach for introductory and pedagogical settings, particularly for audiences with limited coding experience. Such an environment can make digital data construction more accessible, while still acknowledging the procedural limits, infrastructural fragility, and ethical responsibilities that accompany web-based research.

Future work in this area should continue to integrate substantive domain knowledge, cross-source validation, ethical reflection, and computational design. The life and social sciences will increasingly rely on digital sources, but the scientific value of these sources will continue to depend on how carefully they are turned into data.

References

- Megan A. Brown, Andrew Gruen, Gabe Maldoff, Solomon Messing, Zeve Sanderson, and Michael Zimmer. Web scraping for research: Legal, ethical, institutional, and scientific considerations. *Big Data & Society*, 12(1):1–17, 2025. doi: 10.1177/20539517251381686.
- Mine Dogucu and Mine Çetinkaya-Rundel. Web scraping in the statistics and data science curriculum: Challenges and opportunities. *Journal of Statistics and Data Science Education*, 29(2):112–122, 2021. doi: 10.1080/10691898.2020.1787116.
- Charles Ess and Steven Jones. Ethical decision-making and internet research. In *Readings in Virtual Research Ethics: Issues and Controversies*, pages 27–44. Idea Group Publishing, 2004. doi: 10.4018/978-1-59140-152-0.ch002.
- Justin Grimmer, Margaret E. Roberts, and Brandon M. Stewart. *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press, Princeton and Oxford, 2022. ISBN 9780691207544.
- Rob Kitchin. *The Data Revolution: Big Data, Open Data, Data Infrastructures and Their Consequences*. SAGE, London, 2014.
- David Lazer, Alex Pentland, Lada Adamic, Sinan Aral, Albert-László Barabási, Devon Brewer, Nicholas Christakis, Noshir Contractor, James Fowler, Myron Gutmann, Tony Je-

- bara, Gary King, Michael Macy, Deb Roy, and Marshall Van Alstyne. Social science: Computational social science. *Science*, 323(5915):721–723, 2009. doi: 10.1126/science.1167742.
- David M. J. Lazer, Alex Pentland, Duncan J. Watts, Sinan Aral, Susan Athey, Noshir Contractor, Deen Freelon, and Sandra Gonzalez-Bailon. Computational social science: Obstacles and opportunities. *Science*, 369(6507):1060–1062, 2020. doi: 10.1126/science.aaz8170.
- Fumin Li, Yisu Zhou, and Tianji Cai. Trails of data: Three cases for collecting web information for social science research. *Social Science Computer Review*, 39(5):839–855, 2021. doi: 10.1177/0894439319886019.
- Alex Luscombe, Kevin Dick, and Kevin Walby. Algorithmic thinking in the public interest: Navigating technical, legal, and ethical hurdles to web scraping in the social sciences. *Quality & Quantity*, 56(3):1023–1044, 2022. doi: 10.1007/s11135-021-01164-0.
- Ryan Mitchell. *Web Scraping with Python: Collecting More Data from the Modern Web*. O’Reilly Media, 2nd edition, 2018.
- Jakob Ohme, Judith Möller, Wouter van Atteveldt, Sacha Altay, Luca Cardenal, Sebastian Stier, Tabea K. Cornesse, Deen Freelon, Nilam Ram, Byron B. Reeves, and Thomas N. Robinson. Digital trace data collection for social media effects research: APIs, data donation, and screen tracking. *Communication Methods and Measures*, 18(2):124–141, 2024. doi: 10.1080/19312458.2023.2181319.
- José van Dijck, Thomas Poell, and Martijn de Waal. *The Platform Society: Public Values in a Connective World*. Oxford University Press, New York, 2018.